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EarthQuery AI

Problem Statement ID – [Add ID]

- **Problem Statement Title – [Add title]**
- **Theme – [Use PS theme]**
- **PS Category – Software**
- **Team ID – [Add ID]**
- **Team Name – [Portal registered name]**

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EARTHQUERY AI

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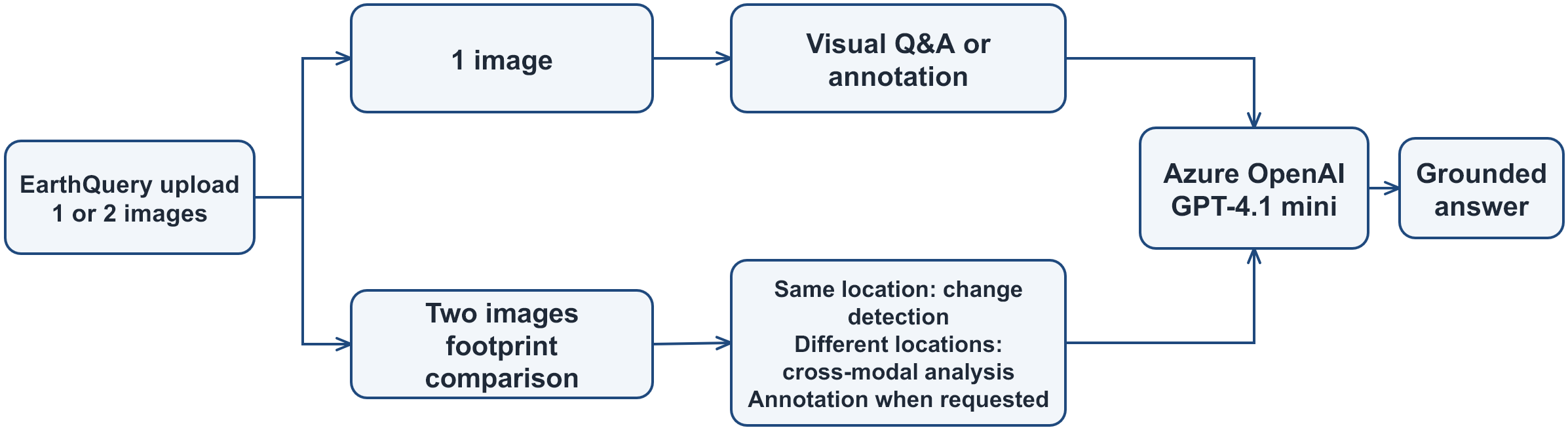
❖ Proposed Solution

- Map-first workstation for natural-language questions about satellite imagery and GeoTIFFs.
- Upload one image for visual question answering or annotation, or two images for change and cross-location analysis.
- Uses raster metadata, geographic context and image evidence to ground each answer.
- Keeps follow-up questions in the same analysis session and displays georeferenced previews on a map.

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TECHNICAL APPROACH

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Stack: Python, FastAPI, Rasterio, Leaflet and Azure OpenAI GPT-4.1 mini

Raster metadata and footprint context join the image evidence and query classification before the model call.

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FEASIBILITY AND VIABILITY

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- Feasible in V6: one or two uploads, raster metadata extraction, georeferenced map placement, and session-based Azure OpenAI Q&A.
- Constraints: files without georeferencing cannot be placed on the map; differing CRS definitions leave footprint relationships unknown. Internet access is needed for basemap tiles and model calls.
- Risks and response: image resolution or ambiguity can limit interpretation. EarthQuery keeps metadata and uncertainty visible so analysts can review answers before acting.

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IMPACT AND BENEFITS

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- Potential users: remote-sensing analysts, researchers, planners and response teams working with raster imagery.
- Social benefit: quicker first-pass review of imagery that may inform situational awareness, with analyst verification retained.
- Economic benefit: less manual effort preparing metadata and comparing image pairs. No measured time or cost savings are claimed.
- Environmental benefit: supports review of land, water and vegetation imagery when suitable data are available.

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RESEARCH AND REFERENCES

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Project source V6 archive: README.md, backend.py, agentic.py, metadata.py, geotiff.py, renderer.py and model.py.
Rasterio Reading datasets documentation
Azure OpenAI Chat completions guide
FastAPI Official tutorial
Leaflet API reference for version 1.9.4